



MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MDS5102/ Segment 01

INTRODUCTION TO PROBABILITY

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Learning Objectives

At the end of this topic, you will be able to:

- Develop sample space for an experiment
- Illustrate real-life examples for various types of events

1. Concepts of Probability

To understand the concept of probability, let us have a look at a few of the important terms:

1. **Experiment:** It is the process of observation that leads to one of the several possible outcomes.
2. **Random experiment:** If the outcome of an experiment is random, it is called a random experiment.
3. **Sample space:** A set of all possible outcomes of an experiment is called the sample space and it is represented by the symbol S .
4. **Event:** An event is a subset of a sample space.

1.1 Example

We can understand these concepts by taking an example.

Experiment: Categorise the cars arriving at a fuel station in a metropolitan city into various types

Sample Space: The car can be a Hatchback, Sedan, MPV, SUV, Crossover, Coupe, Convertible, etc. So, all the possible types of cars in a set are called sample space.

Event: An event is a subset of a sample space. Hence, a car which is not a Hatchback, subspace or the event can be categorised as Sedan, MPV, SUV, Crossover, Coupe, or Convertible.

Note: Depending upon the type of the random experiment, the sample space can be countable or uncountable, i.e., the sample space can have a finite number of elements or an infinite number of elements.

1.2 Intersection of Events

If there are two events, A and B , the intersection of them, denoted by the symbol $A \cap B$, is an event containing all elements that are common to A and B .

Example:

Set $A = \{\text{Students who have cleared examination}\}$

Event $B = \{\text{Student is a female}\}$

So, when A intersects B , it should be the set containing common elements between A and B .

Therefore, $A \cap B = \{\text{Female students who have cleared the examination}\}$

Usually, we use a Venn diagram to denote the intersection of events.

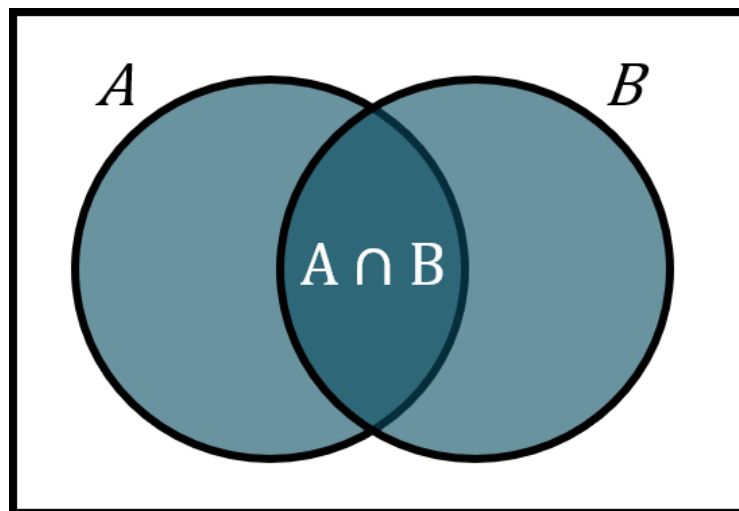


Figure 1: Venn Diagram of Intersection of Events

1.3 Mutually Exclusive Events

The term 'mutually exclusive' means that there is nothing in common. So, two events A and B will be mutually exclusive or disjoint if $A \cap B = \phi$ or $A \cap B$ will not have any elements.

Example:

Let us look into a sample space having some chemical elements.

Sample space $S = \{\text{copper, sodium, nitrogen, potassium, uranium, oxygen, zinc}\}$

Event $A = \{\text{copper, sodium, zinc}\}$ and $B = \{\text{oxygen}\}$

Here, events A and B have no common outcomes, i.e., there are no common elements in the two sets A and B. Therefore, A and B are said to be mutually exclusive as $A \cap B = \{\}$.

The Venn diagram (Figure 2) depicts two sets which do not have any intersections. Thus, they are two different disjoint sets.

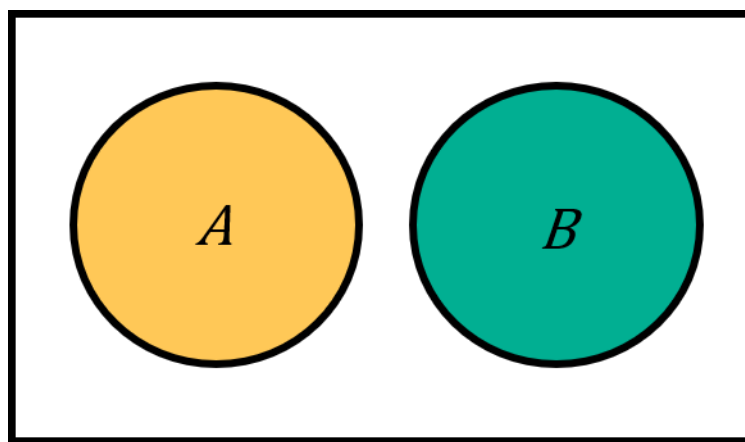


Figure 2: Venn Diagram of Mutually Exclusive Events

1.4 Union of Events

The union of two events A and B , denoted by the symbol $A \cup B$, is the event that contains all the elements that belong to A or B , or both.

Example: Consider

Set $A = \{\text{Cars with sunroof}\}$

$B = \{\text{Cars with diesel engine}\}$

Then $A \cup B = \{\text{Cars with a sunroof or diesel engine}\}$

Thus, a Venn diagram will represent the union of events that contains all those cars which either have a sunroof or a diesel engine, or both. So, this is how we depict $A \cup B$ of two events A and B (shaded region in the Venn diagram below).

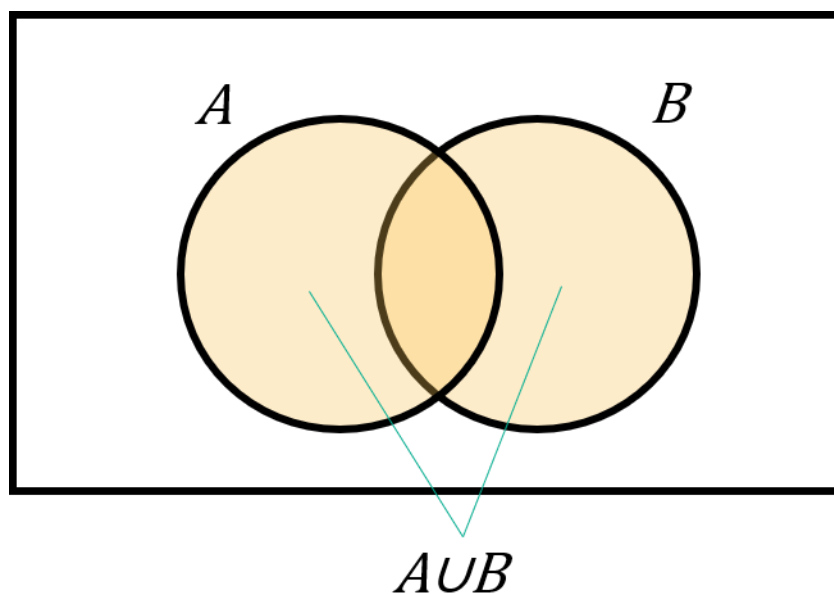


Figure 3: Venn Diagram of Union of Events

1.5 Complementary Events

The complement of an event A with respect to the sample space S is the subset of all elements of S that are not in A .

The complement of A is denoted by the symbol A^c or $\bar{A}$. It can either be written as c or a bar can be put above it, calling it an A complement.

Example:

Let $A = \{\text{Students who have cleared examination}\}$

Then $A^c = \{\text{Students who have not cleared examination}\}$

So, whatever is not there in A , but present in the sample space S , happens to be the complement of A .

Similarly, it can be represented in the Venn diagram.

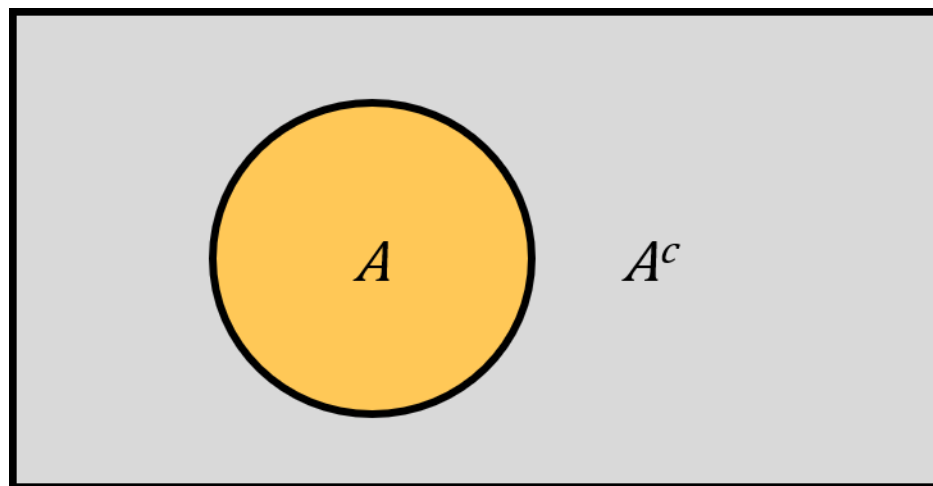


Figure 4: Venn Diagram of Complementary Events

1.6 Independent Events

Two events, A and B, are said to be independent if the occurrence of one does not affect the occurrence or the non-occurrence of the other event.

Example:

Let there be two events,

A = {Pet owner}

B = {Post-graduate}

Owning a pet or having a postgraduate degree are not related to each other. Therefore, A and B are independent events as owning a pet is not related to an individual's educational status. So, independent events can be identified using probability statements.

1.7 Conditional Probability

The most common method of identifying whether two events are independent or not depends on a type of probability called conditional probability.

Suppose there are two events A and B, each with a positive probability of occurrence, then the probability of occurrence of A, given that B has already occurred, is denoted by

$$P(A \mid B) = (P(A \cap B)) / (P(B)).$$

This is called conditional probability. Similarly, the probability that B occurs, given that A has occurred (B conditioned on A) is equal to

$$P(B \mid A) = (P(A \cap B)) / (P(A)).$$

1.7.1 Independent Versus Dependent Events

The conditional probabilities are required to identify whether events A and B are independent or not. If two events A and B are independent, then the probability of A occurring—given B has already occurred—should not be affecting the occurrence of A.

Hence, the probability of $P(A|B) = P(A)$, or equivalently $P(B|A) = P(B)$. In this case, A and B will be independent of each other.

Otherwise, if these relationships are not there, then the given events A and B are said to be dependent.

2. Probability: Rules

There are a few rules to follow in probability.

2.1 The Complement Rule

The complement rule states that if A and A^c are complement of each other, then

$$P(A) + P(A^c) = 1$$

because the total probability is always 1 and hence, the probability of the sample space is 1. It further, can be written as

$$P(A^c) = 1 - P(A).$$

This is the first rule known as the complement rule.

2.2 The Addition Rule

To check the probability of A or B occurring, given there are two events A and B, the term union for A or B is used. So, it will be a probability of A union B, which can be spelt as the probability that A or B occurs.

The addition rule states that the probability that A or B occurs, or that at least one of these events occurs, is equal to the probability that A occurs, plus the probability that B occurs, minus the probability that both A and B occur, or equivalently

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

2.2.1 The Addition Rule for Mutually Exclusive Events

When there are mutually exclusive events, the addition rule reduces to

$$P(A \cup B) = P(A) + P(B).$$

For non-mutually exclusive events, the addition rule is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Only in the case of mutually exclusive events, the last term $P(A \cap B) = 0$.

2.3 The Multiplication Rule

The probability of A intersection B will be equal to the probability of the occurrence of B into the probability of the occurrence of A, given B has already occurred.

$$P(A \cap B) = P(A|B)P(B).$$

Or equivalently, there is a probability that A has occurred into a probability that B has occurred, given that A has already occurred.

$$P(A \cap B) = P(B|A)P(A).$$

2.3.1 The Multiplication Rule for Independent Events

The multiplication rule for independent events states that if A and B are independent, the probability of A and B equals the product of the unconditional probabilities of A and B, i.e.,

$$P(A \cap B) = P(A)P(B).$$

Similarly, the probability of A occurring given B has occurred is the probability of B alone.

Therefore,

$$P(A \cap B) = P(A)P(B)$$

if and only if A and B are independent of each other.

3. Summary

In this topic, we discussed:

- Basic concepts of probability such as random experiment, sample space, and events
- Different types of events such as mutually exclusive events, complimentary events, and independent events



References:

- Ross, S M. (2014). *Introduction to probability models*. Academic Press.
- Brownlee, J. (2019). *Probability for machine learning: Discover how to harness uncertainty with Python*. Machine Learning Mastery.



E-references:

- <https://probabilitycourse.com/>